

Supplementary Experimental Report:
Raw-Frame Analysis, Anchor Lower-Trim Experiment,
and Phase Center Sensitivity

Erlangen 28-May-2026 V5 Campaign – Additional Analysis

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Contents

1	Context	2
2	Phase Center Sensitivity Analysis	2
3	Raw-Frame Range Distribution Analysis	3
4	Anchor Lower-Trim Blind Experiment	5
5	Solver Audit Summary	6
6	Updated Headline Table	7
7	Updated Claim Matrix	8
8	Engineering Recommendations	8
9	Open Questions	9

1 Context

This document supplements the original English Vicor validation report and records the experiments that were completed after that report was sent for review. The original campaign left a central paradox: the V5 common-mode anchor layout corrected the anchor-side metric scale, with Sim3 scale moving from 0.958 for V4 to 1.010 for V5, yet the older V4 layout still gave the lower p50 static median on the same 24-position Erlangen campaign. In the transfer matrix, V4+C_V4+D_LOO gives 57.9 mm median 3D error, while V5+C_V5+D_LOO gives 67.8 mm. Those values are reproduced from `FULL_V5_scale_to_vicon/tables/v5_vs_v4_scale_comparison.csv` and `FULL_transfer_matrix/tables/transfer_matrix_48cells.csv`.

The additional experiments were designed to test whether this paradox is caused by delay–layout coupling rather than by a simple V4/V5 implementation error. The phase-center sensitivity analysis asks whether Vicor marker offsets can overturn the ranking. The raw-frame campaign uses the stored per-frame tag-anchor distributions to reduce positive NLOS tails. The anchor lower-trim blind experiment asks whether the same lower-tail statistic should also be used for inter-anchor self-calibration. The solver audit records which parts of the V1–V5 and T1–T4 solvers are actually algorithmic changes and which parts are preprocessing or policy changes.

The main conclusion is narrower and stronger than an accuracy claim. V5 is the physically better anchor geometry by metric scale, while V4 is the p50 single-environment positioning winner. When the positive tag-anchor range tail is reduced in static batch mode, the empirical ranking flips: V5 reaches 44.5 mm LOO, whereas the best V4 geometry row in the same raw-frame ranking is 53.9 mm. This flip is the strongest causal evidence that V4’s p50 advantage came from beneficial cancellation between a distorted layout and structured positive range bias.

2 Phase Center Sensitivity Analysis

The phase-center sensitivity batch addressed a necessary alternative explanation. Vicor marker centers are not guaranteed to coincide with UWB antenna phase centers, and a systematic marker-to-antenna offset could make a geometrically correct UWB layout appear worse against the optical reference. The analysis therefore perturbed phase centers and repeated the relevant ranking checks rather than assuming that the marker geometry was exact.

The global shift sweep found that the V4-over-V5 p50 static ranking does not flip for tested offsets up to more than 10 mm. The manufacturing-variation Monte Carlo used 5000 samples per sigma level and found that $P(\text{V4 beats V5})$ remains at least 0.998 through $\sigma = 8$ mm. The Vicor oracle rank is less stable: the table reports a fragile threshold around 2 mm for the oracle rank or worst-status conclusion. The cancellation valley itself remained invariant to the tested phase-center offsets, which means that the delay–layout coupling result is not an artifact of one exact optical marker model.

Table 1: Phase-center robustness summary. Values are reproduced from FULL_V5_phase_center_sensitivity/tables/a6_robustness_summary.csv.

Conclusion	Baseline value	Flip threshold	Robustness
V5 Sim3 scale > 0.99	1.010	>10	robust
V4+LOO beats V5+LOO	V4-V5=-23.0 mm	>10	robust
Vicon oracle rank/worst status	rank=2, worst=False	2.0	fragile
D_tag LOO approximately 49.6mm	49.028 mm; sensitivity 0.190 mm/mm	not binary	stable
D_tag per-height spread V5 < V4	7.4 < 11.8 mm from prior mechanism audit	not directly flipped by global phase-center sweep	not directly tested here; use A4 as caveat
Cancellation valley exists	max tested operating-point valley-distance shift 11.37	does not depend on absolute phase-center offset	invariant mechanism

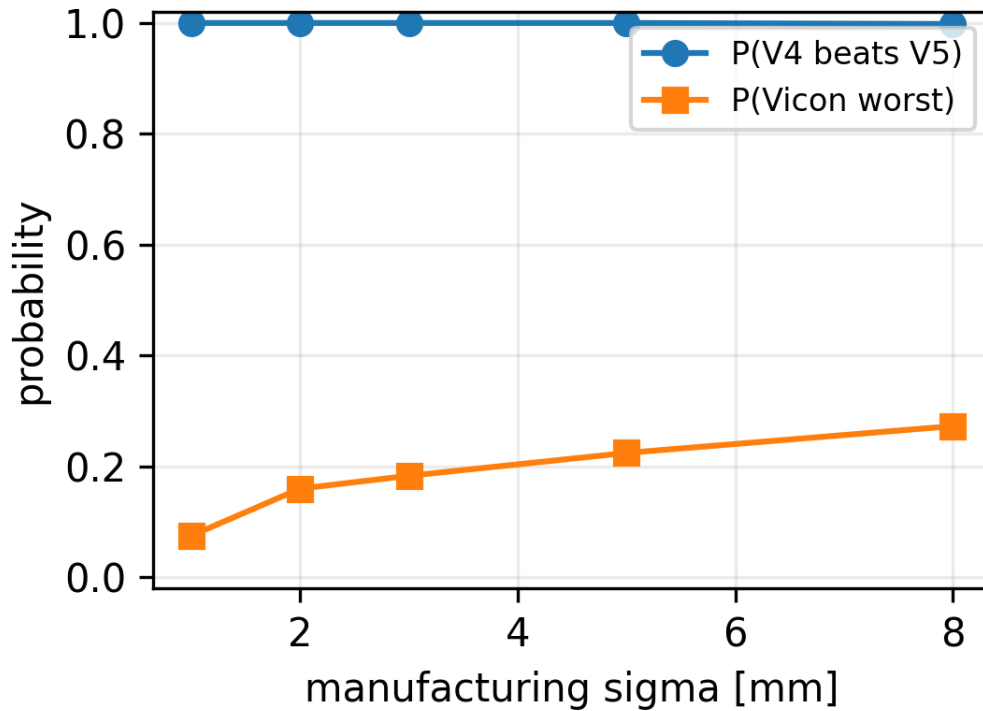


Figure 1: Manufacturing-variation phase-center Monte Carlo. The static V4-over-V5 p50 ranking is robust across the tested perturbation range, while the Vicon oracle rank is more fragile. Source: FULL_V5_phase_center_sensitivity/figures/a2_ranking_probability_vs_sigma.png.

3 Raw-Frame Range Distribution Analysis

The raw-frame campaign began with a data-availability check. The static captures contain the raw per-frame tag-anchor observations needed for distribution-level analysis: 230,544 raw rows, 228,265 valid rows, and approximately 1200 frames per position-anchor link. This matters because the historical p50 pipeline collapses each link histogram to one median value before solving. If the histogram has a positive NLOS tail, a lower-quantile statistic can move the measurement closer to the LOS-side component without using CIR data.

Table 2: Raw-frame campaign key numbers. Values are from FULL_V5_experimental_report/tables/rawframe_v3_key_results.csv.

Metric	Value	Unit
Raw rows	230544.0	rows
Valid rows	228265.0	rows
Full grid configurations	107448.0	rows
Best honest LOO median	44.5	mm
Oracle lower bound	44.6	mm
Bootstrap CI	[33.4, 82.8]	mm

The important result is the ranking flip. Under p50 aggregation, V4 wins the static LOO comparison despite having the wrong metric scale. Under lower-quantile static batch aggregation, V5 becomes the best geometry: lower_trim_20 with V5 and Huber30 gives 44.5 mm median LOO, while the best V4 geometry row in the same honest raw-frame ranking is 53.9 mm and the best Vicon geometry row is 44.7 mm. The oracle lower bound in the v3 ladder is 44.6 mm, so the V5 lower-tail row nearly reaches the recoverable median implied by the available raw distributions.

This is not presented as a new NLOS detection method. The lower-tail statistic is deliberately simple. Its value is experimental: it reduces the positive tag-anchor tail enough to remove the cancellation condition under which V4 was winning. The intervention therefore supports the mechanism claim that single-environment positioning accuracy can prefer a physically wrong geometry when the wrong geometry cancels structured range bias.

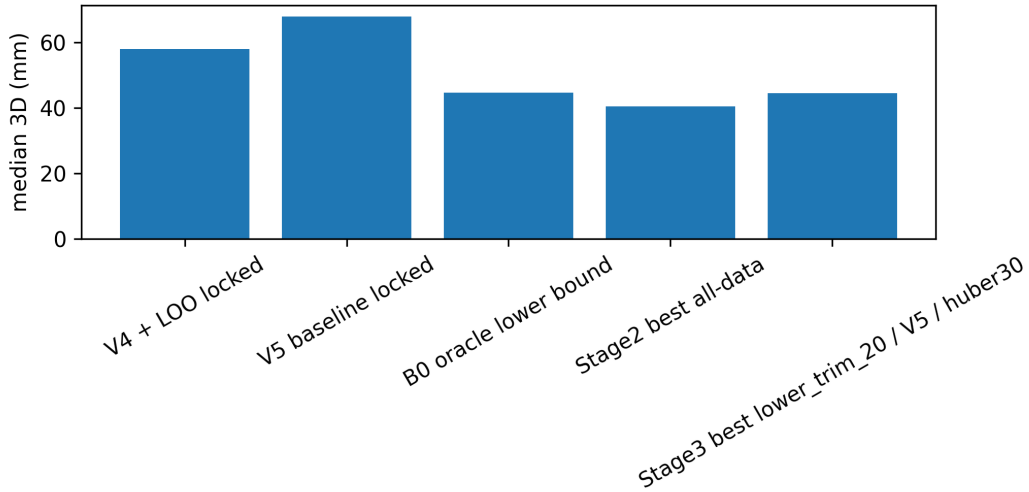


Figure 2: Raw-frame accuracy ladder. The lower-quantile V5 static batch row approaches the oracle median, while the tail remains unresolved. Source: FULL_V5_rawframe_bruteforce_v3/figures/s6_accuracy_ladder.png.

The caveat is the tail. The lower-quantile V5 row improves the median to 44.5 mm but keeps P95 at 164.1 mm and RMSE at 81.5 mm. By contrast, V4+C_V4+D_LOO under p50 has P95 110.6 mm. The lower-tail result therefore solves the typical static error better than it solves worst-case static positioning. The 5000-iteration bootstrap interval, 33.4–82.8 mm, also remains wide because the campaign has only 24 static positions.

4 Anchor Lower-Trim Blind Experiment

The raw-frame tag-anchor result raised a second question: if lower-tail statistics recover a better LOS-side tag-anchor range, should the same statistic be used for inter-anchor self-calibration? The blind anchor lower-trim experiment tested this directly by rebuilding V5-style anchor layouts with several inter-anchor aggregation methods and `e_i` settings, then replaying the tag positioning pipeline.

The answer was negative. The inter-anchor distributions contain 56,000 valid rows across 28 pairs, with a median of 2000 frames per pair. Their mean skewness is 0.063, which is close to symmetric, while the mean `p50` minus `lower_trim_20` difference is 32.9 mm. In other words, `lower_trim_20` systematically shortens the inter-anchor distances even though those distributions do not show the same right-tail structure as the static tag-anchor histograms.

Table 3: Anchor lower-trim blind-experiment key numbers. Values are from `FULL_V5_experimental_report/tables/anchor_lower_trim_key_results.csv`.

Metric	Value	Unit
AA valid rows	56000.0	rows
Median frames per pair	2000.0	frames
Mean skewness	0.063	skewness
Mean <code>p50</code> - <code>lower_trim_20</code>	32.9	mm
Best <code>lower_trim_20</code> anchor row	46.4	mm LOO
Current <code>p50</code> control	44.5	mm LOO
Best <code>p50</code> <code>e_i=0</code> row	43.2	mm LOO
<code>P(new wins), e_i=0</code>	0.659	probability

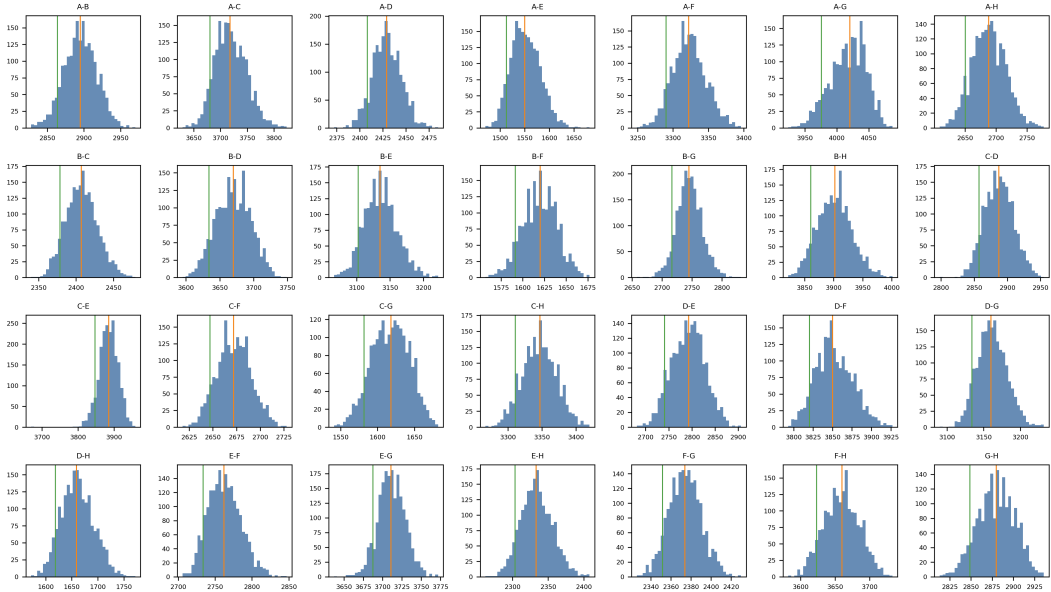


Figure 3: Inter-anchor raw range histograms. The near-symmetric distributions explain why lower-tail inter-anchor aggregation introduces a downward range bias rather than removing a positive tail. Source: `FULL_V5_anchor_lower_trim/figures/l1_inter_anchor_histograms.png`.

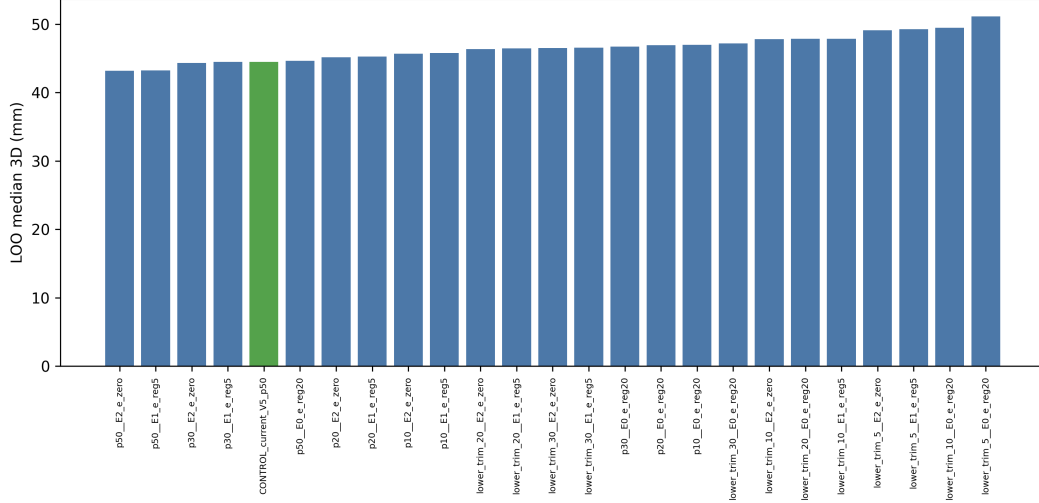


Figure 4: Blind anchor-aggregation comparison. The best lower_trim_20-anchor row is worse than the p50 control, while p50 with e_i=0 is a small, not yet statistically locked, candidate improvement. Source: FULL_V5_anchor_lower_trim/figures/13_accuracy_by_anchor_method.png.

The practical rule from this experiment is simple: keep p50/median for inter-anchor self-calibration in this dataset, and treat lower-quantile aggregation as a tag-anchor static batch estimator. The e_i result is separate. p50 with e_i=0 reaches 43.2 mm LOO, compared with 44.5 mm for the current p50/e_reg20 control, but the paired bootstrap gives $P(\text{new wins}) = 0.659$ and a confidence interval crossing zero. This is a candidate setting for the next capture, not a locked replacement.

5 Solver Audit Summary

The solver audit found five core anchor-layout solver generations and four C-level tag solver versions, plus one Python tag-solver policy variant. The audit matters because several campaign results are preprocessing or parameterization effects rather than new numerical solvers. In particular, no production T5 tag solver was found; the lower-trim and Huber30 improvements are changes to range aggregation and loss configuration around the existing tag-solver family.

Table 4: Anchor solver evolution from the read-only audit. Source: ../../../../Analysis/solver_audit/reports/SOLVER_AUDIT_SUMMARY.md.

Version	One-line description	Delay model or key parameter
V1 / v1-old	early MDS baseline	no delay
V2	inverse-variance bidirectional fusion	no delay
V3-lite	median/MAD/MVUE fusion	no delay
V3-full	robust fusion plus alternating per-anchor delays	per-anchor delays
V4 / v4-io	bounded independent-delay Huber inter-anchor solve	dA=0, [-60,+60] mm
V5 / common-mode	common-mode extension, di=c+ei	c=111.985 mm, e_reg=20 mm artifact

Table 5: Tag solver evolution from the read-only audit. Source: `../../../../Analysis/solver_audit/reports/SOLVER_AUDIT_SUMMARY.md`.

Version	One-line description	Status
T1	robust WLS multilateration	current method default
T2	T1 plus quality-aware sigma inflation	C method 2
T3	T2 plus residual EMA and temporal prior	C method 3
T4	adaptive full-anchor T1 / low-redundancy T3 policy	Python wrapper plus C method 4
T4_V6_IMU_GATH4	T4 plus accelerometer-gated temporal prior	policy variant

The current research configuration is V5 common-mode self-calibration with the existing artifact at $c = 111.985$ mm and `e_reg20`, combined with a tag solver that defaults to robust WLS and Huber scale 30 mm after recent edits. The audit recommends keeping `p50` inter-anchor aggregation, testing lower-tail tag-side aggregation for static batch operation, explicitly labeling `e_i` settings, and fixing the C-core loss-selector mismatch where the Python interface exposes a linear option but C validation resets it to Huber.

6 Updated Headline Table

The headline table below is reproduced from `FULL_V5_experimental_report/tables/locked_headline_v2.csv`. It should be read as a campaign registry rather than a single leaderboard, because rows differ in evaluation protocol. Rows labelled in-sample, diagnostic, Sim3, or oracle should not be used as deployment claims. Row O is the new raw-frame static batch result; its median is strong, but its P95 is not.

Table 6: Updated locked headline table. Empty cells indicate metrics not reported for that row in the source table.

Row	Variant	Median	P95	RMSE	Evaluation	New
A	V4 production	71.9	176.0	110.4	in-sample, all 24	
B	V4 + D_LOO	57.9	110.6	74.4	LOO-CV	
C	V5 baseline	67.8	160.5	86.4	LOO-CV	
D	V5 apparent best	56.0	143.1	79.5	in-sample post-selected	
E	V4 apparent best	54.9	154.8	79.6	in-sample post-selected	
F	V5 corrected	65.6			OOB-bootstrap	
G	V4 corrected	64.5			OOB-bootstrap	
H	V5 bootstrap CI	[54.3, 63.7]			bootstrap 95% CI	
I	Nested CV (height)	82.9			held-out test	
J	Nested CV (quadrant)	88.0			held-out test	
K	Nested CV (spatial6)	94.2			held-out test	
L	ROTO V5 per-frame	101.5	214.4	126.2	BEST-FIT-ALIGNED	
M	ROTO SE(3) aligned	82.5	185.2	103.7	diagnostic	
N	ROTO Sim3 aligned	74.3	160.8	94.8	diagnostic only	
O	lower_trim_20 + Huber30 + V5	44.5	164.1	81.5	LOO-CV	YES
P	lower_trim_20 + Huber30 + V5(<code>e_i=0</code> anchor refit)	43.2	163.1	81.8	LOO-CV; anchor refit diagnostic	YES
Q	Oracle lower bound	44.6			oracle	YES
R	Bootstrap (lower_trim_20) CI	[33.4, 82.8]			bootstrap 95% CI	YES

The most important comparison is not Row O against every other row, but the ranking reversal under a controlled change in tag-anchor range bias. Under `p50`, V4 wins at 57.9 mm versus V5 at 67.8 mm. Under `lower_trim_20` with Huber30, V5 reaches 44.5 mm and the best V4 raw-frame geometry row is 53.9 mm. That reversal is the evidence for delay–layout–NLOS coupling.

7 Updated Claim Matrix

The V3 report reorganizes the claim matrix around delay–layout coupling. The campaign-level counts are 10 Level A claims, 10 Level B claims, 5 Level C claims, and 2 Level D claims after removing the mixture-model ranking as a paper-level scientific claim and downgrading the lower-quantile method claim to a caveated engineering result. The full claim register remains in `FULL_V5_experimental_report/tables/claim_evidence_matrix_v2.csv`; the V3 narrative is the controlling interpretation where the prose and the raw CSV differ.

Table 7: New or revised claims introduced by the raw-frame and anchor lower-trim work.

ID	Level	Recommended wording
26	B	Lower-quantile tag-anchor aggregation improves static median when enough frames are accumulated, but the P95 remains high.
27	A	Inter-anchor raw ranges are nearly symmetric; lower-tail aggregation is inappropriate for anchor self-calibration here.
28	B	The $e_i=0$ or low- e_i V5 variant is slightly better for the raw-frame static estimator, but the bootstrap evidence is not decisive.
29	A	When tag-anchor positive bias is reduced, V4’s cancellation advantage disappears and V5 becomes the better geometry.

The practical wording should stay conservative. `Lower_trim_20` is a static batch extractor for this campaign, not a CIR-free NLOS detector. V5 transfer to new rooms remains untested. ROTO still has a dynamic floor around 101.5 mm under the conservative BEST-FIT-ALIGNED convention. The central paper claim is the mechanism: a physically wrong self-calibration can win in the same environment because its scale error cancels structured positive range bias.

8 Engineering Recommendations

For anchor self-calibration, use p50/median inter-anchor ranges in Erlangen-style captures. The inter-anchor distributions are nearly symmetric and lower-tail aggregation worsened the blind replay. The V5 common-mode family should be reported with its e_i setting. The $e_i=0$ and e_{reg5} variants are worth testing in the next campaign, but e_{reg20} should not be silently replaced on the basis of a 24-position bootstrap with $P(\text{new wins}) = 0.659$.

For static tag solving, keep raw per-frame ranges and test lower-quantile aggregation when enough frames per link are accumulated. In this campaign, the useful row is “V5 p50 anchors, `lower_trim_20` tag ranges, Huber30, `D_tag` LOO.” The `D_tag` value is estimator-specific: the p50 V5 LOO value is 49.621 mm, whereas the lower-quantile V5/Huber30 row calibrates around a much smaller effective tag delay. These numbers should not be mixed.

For dynamic operation, do not transfer the static lower-tail result directly to one-frame ROTO solves. ROTO remains a separate limitation, and the current conservative headline is 101.5 mm BEST-FIT-ALIGNED. The alignment audit and time-offset sweep should be reported as diagnostics, not as a solved dynamic-tracking pipeline.

For firmware and capture design, the priority is to keep saving raw per-frame ranges. The raw data already made the ranking-flip experiment possible. A future quasi-static mode can accumulate a short histogram only when IMU and range stability indicate that the tag is not moving, then compare p50 and lower-quantile solves within that window.

9 Open Questions

The first open question is tail reduction. The lower-quantile row improves the static median but leaves P95 around 164 mm. The next analysis should identify which positions and anchors cause this tail and whether per-anchor stability rules can reduce it without giving back the median gain.

The second open question is external validation. The current LOO and bootstrap results are still within one room and one 24-position campaign. A second-room run with a frozen pipeline should repeat p50 V4, p50 V5, lower-quantile V5, p50/e_i=0 anchors, and the same hard validation splits.

The third open question is D_{tag} self-calibration without known positions. The present D_{tag} values depend on Vicon-supported static folds or known-position residuals. Deployment needs a small-calibration procedure that can estimate the appropriate device- and estimator-specific D_{tag} without optical ground truth.

The fourth open question is dynamic transfer. Static lower-quantile aggregation requires a histogram, while ROTO has single-frame measurements. The next dynamic work should test quasi-static sliding windows, hardware time synchronization, and physical tag-orientation sweeps before claiming a real-time dynamic improvement.